

Kenya Inflation Risk Audit (1960–2024)

A Statistical and Policy Analysis Report

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CONFIDENTIAL MEMORANDUM

TO: The Governor, Central Bank of Kenya

FROM: Stephen Munene (Policy Analysis Unit)

DATE: March 21, 2026

SUBJECT: Retrospective Analysis of Inflation Volatility (1960–2024): A Statistical Risk Audit

1. Executive Summary

This memorandum provides a quantitative "health check" of Kenya's price stability over the last 65 years. By applying a **Z-score framework**—a method of measuring how "abnormal" a year is relative to history—we have identified the structural DNA of Kenyan inflation.

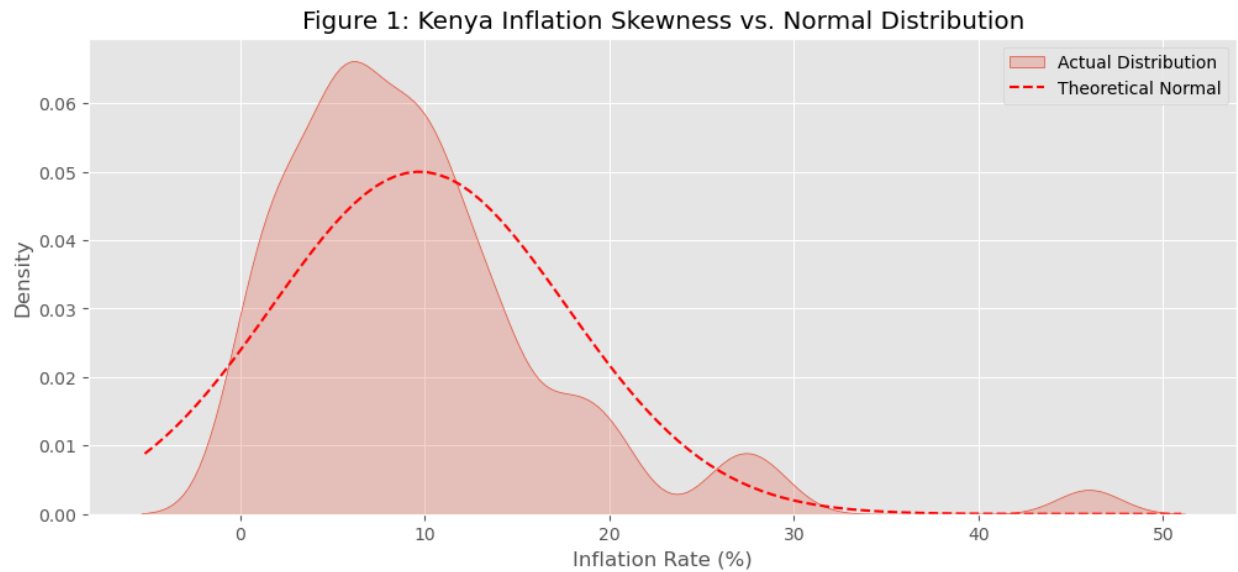
Our findings indicate that while the CBK has successfully anchored inflation in recent years, the historical "DNA" of the Shilling is characterized by high volatility and a tendency for sudden, extreme upward shocks (Right-Skewness).

2. The "DNA" of Kenyan Inflation

To understand where we are going, we must define our historical baseline.

- **The Anchor (Mean):** Since 1960, Kenya's average annual inflation has been **9.7%**.

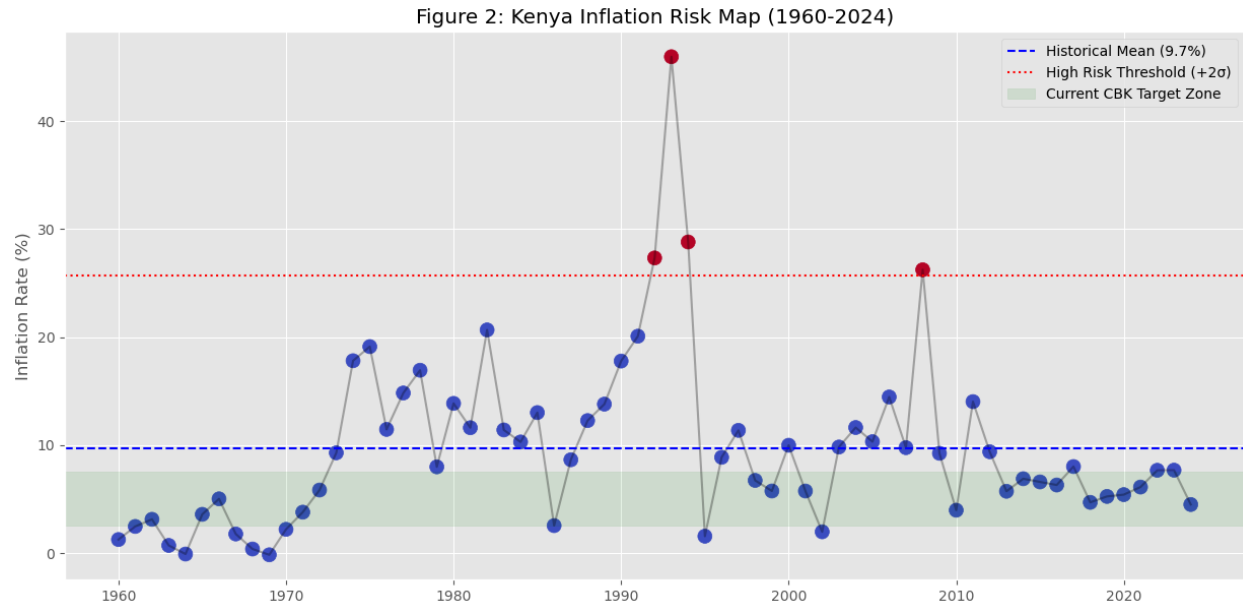
- **The Volatility (Standard Deviation):** Our "swing factor" is **7.98%**. In practical terms, this means that in any given year, it is "statistically normal" for inflation to fluctuate wildly between 1.7% and 17.7%.



3. Identification of "Black Swan" Events (Risk Audit)

We utilized **Z-scores** to flag years that were mathematically "Extreme." A Z-score of +2.0 or higher represents a crisis that occurs less than 5% of the time.

- **The 1993 Hyper-Shock ($Z = 4.54$):** This remains the greatest outlier in our recorded history. With inflation at 45.9%, this was a "4-Sigma" event—statistically, something that should almost never happen. It serves as a reminder of the risks of rapid money supply expansion.
- **The 2008 Supply Shock ($Z = 2.07$):** The post-election crisis and global oil spike pushed us into the "High Risk" zone, proving how external shocks can break our internal stability.



4. Policy Implications & Realism

The current CBK target of **5.0% ± 2.5%** is an ambitious departure from our 65-year history.

1. **The "Efficiency" Gap:** Our current target (max 7.5%) is significantly lower than our historical average (9.7%). This confirms that the modern CBK is far more "hawkish" and disciplined than the administrations of the 70s and 90s.
2. **Buffer Requirements:** Because our volatility (σ) is high, the Bank must maintain robust foreign exchange reserves to defend the Shilling against the "Right-Tail" risks identified in this report.

5. Conclusion

Kenya's inflation is not a "Normal Distribution." It is a "Shock-Prone Distribution." While the last decade shows a tightening of control, the ghost of the 1993 Z-score (4.54) suggests that price stability is a fragile equilibrium that requires constant, data-driven vigilance.

Recommendation: The MPC should continue to treat any Z-score move toward +1.5 as an immediate signal for intervention before it cascades into a "2-Sigma" crisis.

Stephen Munene,
Lead Data Scientist.

TECHNICAL APPENDIX

Statistical Methodology and Data Analysis

Purpose of Appendix:

This section provides the statistical foundation supporting the policy conclusions presented in the memorandum. All findings are reproducible and derived using standard quantitative methods.

Technical Appendix A: Inflation Volatility & Risk Analysis

Prepared for: Policy Simulation (Central Banking Context)

Analyst: Stephen Munene

Date: March 2026

1. Business & Economic Understanding

The primary mandate of the Central Bank of Kenya (CBK) is to maintain price stability.

Currently, the target inflation range is $\pm 2.5\%$.

Objective: This notebook performs a statistical audit of Kenya's inflation history (1960-2024). We use "Data DNA" (Mean, Standard Deviation, Z-scores) to:

1. Identify the "Black Swan" events (High-risk anomalies).

2. Determine if the distribution is "Normal" or "Fat-Tailed" (skewed by extreme shocks).
3. Provide a quantitative baseline for future Monetary Policy Committee (MPC) decisions.

2. Data Preparation & Environment

We use standard quantitative stacks. We apply the **ggplot** style for clean, academic grade visualizations.

In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import norm

# Set aesthetic parameters for a professional report
plt.style.use('ggplot')
plt.rcParams['figure.figsize'] = (12, 6)

# Loading the dataset
df = pd.read_csv('kenya_inflation_data.csv')
```

3. Exploratory Data Analysis(EDA) & Data DNA

Before modeling we must understand the "DNA" of our currency's purchasing power.

In [2]:

```
# Calculating the 'DNA' metrics
mu = df["inflation"].mean()
sigma = df["inflation"].std()

print(f"Historical Mean ( $\mu$ ): {mu:.2f}%")
print(f"Historical Volatility ( $\sigma$ ): {sigma:.2f}%")
Historical Mean ( $\mu$ ): 9.70%
Historical Volatility ( $\sigma$ ): 7.98%
```

Analytical Note:

A Standard Deviation of 7.98% against a Mean of 9.70% indicates **High Volatility**. In a stable economy, we would expect σ to be much lower (2–3%). This suggests Kenya's inflation is prone to external and internal shocks (droughts, oil prices, policy shifts)

4. Risk Analysis: The Z-Score Framework

We use the Z-score to transform raw percentages into "Units of Risk".

- **Z > 2:** Extreme Inflationary Pressure (Top 2.5% of historical cases).
- **Z < - 2:** Significant Deflationary Pressure(Rare in Kenya).

In [3]:

```
# Feature Engineering: Z-Score and Risk Categorization
df['z_score'] = (df['inflation'] - mu) / sigma

# Define Risk Flags based on 2-Sigma Thresholds
df["risk_status"] = df["z_score"].apply(
    lambda x: "Extreme Outlier" if abs(x) > 2 else "Manageable"
)

# Identifying the "Crisis Years"
crises = df[df["risk_status"] == "Extreme Outlier"]
display(crises[['year', 'inflation', 'z_score']])
```

	year	inflation	z_score
32	1992	27.332364	2.209214
33	1993	45.978881	4.545823
34	1994	28.814389	2.394928
48	2008	26.239817	2.072306

In []:

5. Visualizing the Distribution (The "Policy Gap")

As an analyst, I need to see if the data follows a **Normal Distribution**. If it doesn't, standard "avarage" policy won't work.

In [4]:

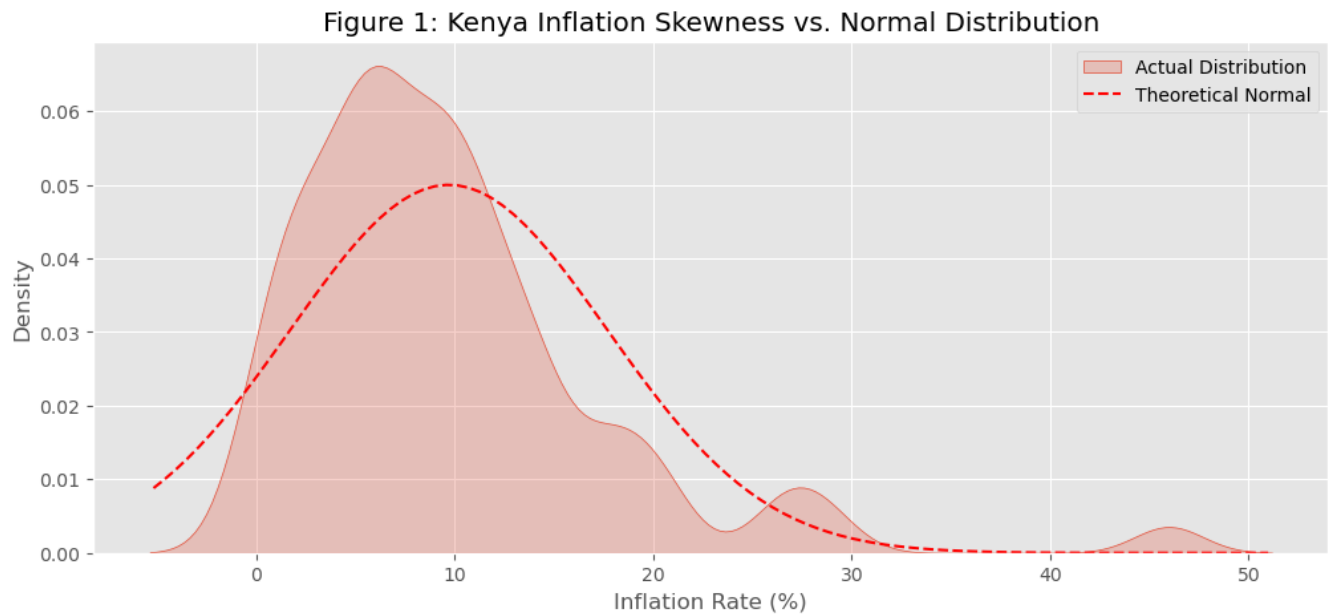
```
plt.figure(figsize=(12, 5))

# Plot 1: The Probability Density
sns.kdeplot(df['inflation'], bw_adjust=0.5, fill=True, label="Actual
Distribution")

# Plot 2: The Theoretical Normal Curve for comparison
x = np.linspace(df['inflation'].min() - 5, df['inflation'].max() + 5, 100)
plt.plot(x, norm.pdf(x, mu, sigma), 'r--', label="Theoretical Normal")

plt.title("Figure 1: Kenya Inflation Skewness vs. Normal Distribution")
```

```
plt.xlabel("Inflation Rate (%)")
plt.legend()
plt.show()
```



Insight for the Governor:

Our distribution is **Positively skewed** (Right-skewed). This means Kenya is more likely to experience "Hyper-Inflationary Shocks" (like the 45% in 1993) than it is to experience equivalent deflation. Policy must therefore be structurally biased toward "Hawkish" inflation containment.

6. Historical Time-Series Risk Map

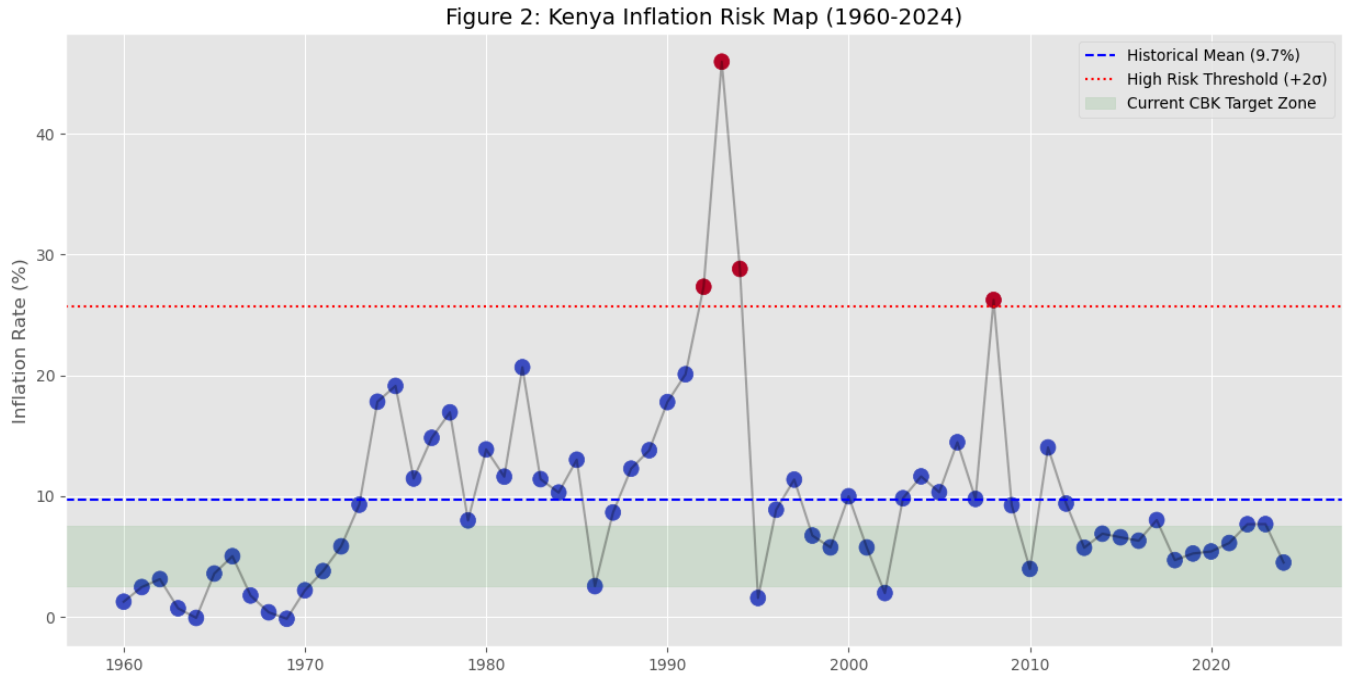
This the "Money Shot" for the memorandum. It shows exactly when the economy broke the 2-Sigma barrier.

In [5]:

```
plt.figure(figsize=(15, 7))
plt.plot(df["year"], df["inflation"], color='black', alpha=0.3)
plt.scatter(df["year"], df["inflation"], c=(abs(df['z_score']) > 2),
            cmap='coolwarm', s=100)

# Critical Policy Lines
plt.axhline(mu, color='blue', linestyle='--', label=f'Historical Mean ({mu:.1f}%)')
plt.axhline(mu + 2*sigma, color='red', linestyle=':', label='High Risk Threshold (+2σ)')
plt.axhspan(2.5, 7.5, color='green', alpha=0.1, label='Current CBK Target Zone')
```

```
plt.title("Figure 2: Kenya Inflation Risk Map (1960-2024)")
plt.ylabel("Inflation Rate (%)")
plt.legend()
plt.show()
```



7. Final Recommendations for the Memorandum

Based on the notebook Analysis:

1. Anomaly Detection: 1993 remains the most significant outlier in Kenya's history (\$Z \approx 4.5\$).

2. Volatility Warning: Because our Standard Deviation (7.98) is nearly as large as our Mean (9.7), the CBK must maintain higher-than-average reserves to buffer against "2-Sigma" shocks.

3. Target Realism: The current target of 7.5% is significantly below the 65-year historical mean, suggesting a modern era of much tighter monetary discipline compared to the 20th century.

DISCLAIMER

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